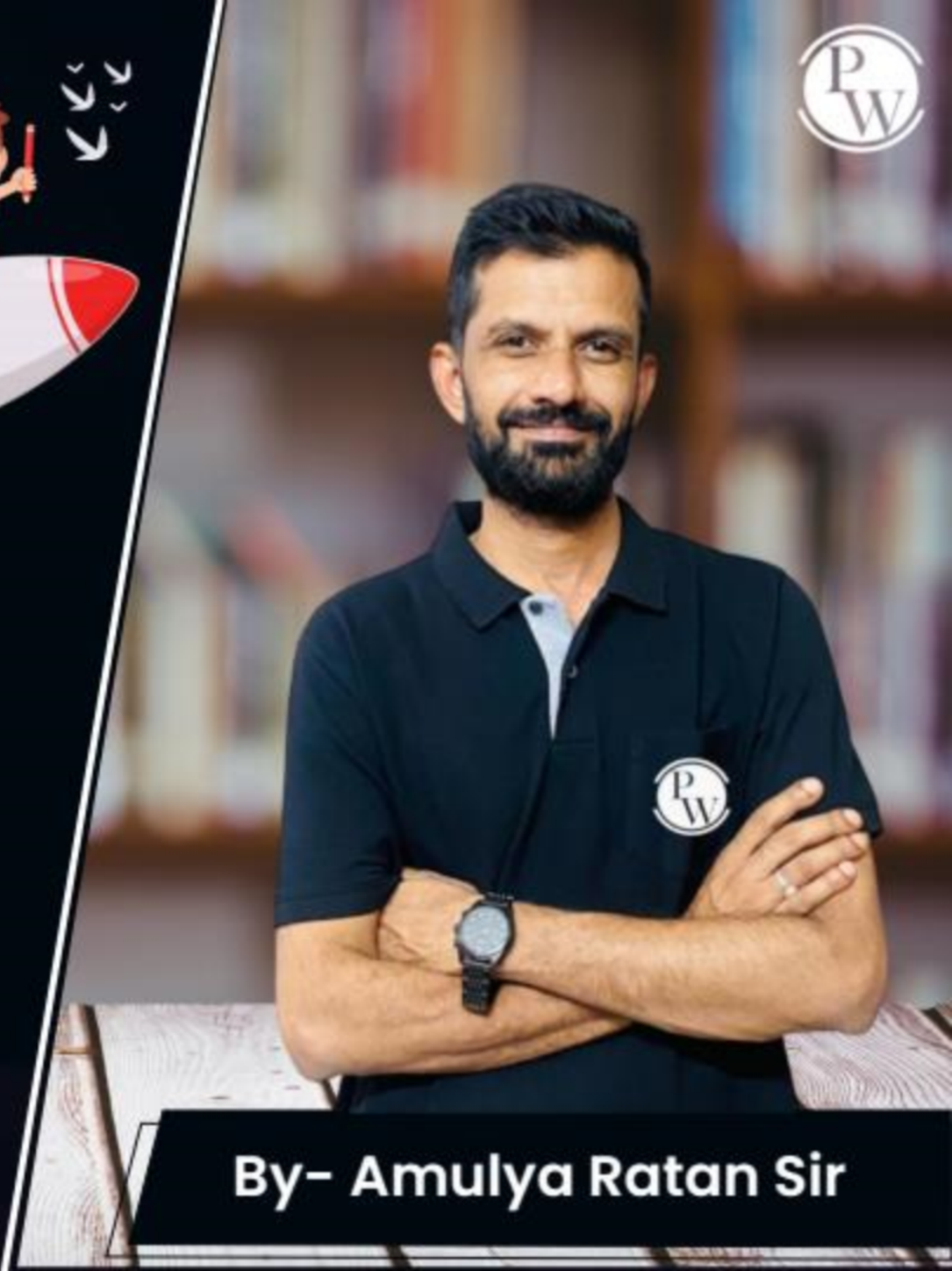
GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 07



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic Profit Loss and Puzzle

Topics to be Covered



Topic-1 Concept of Mixtures Alligations

Topic-2 Understanding basics of Numbers

Topic-3 Questionnaire

[MCQ]

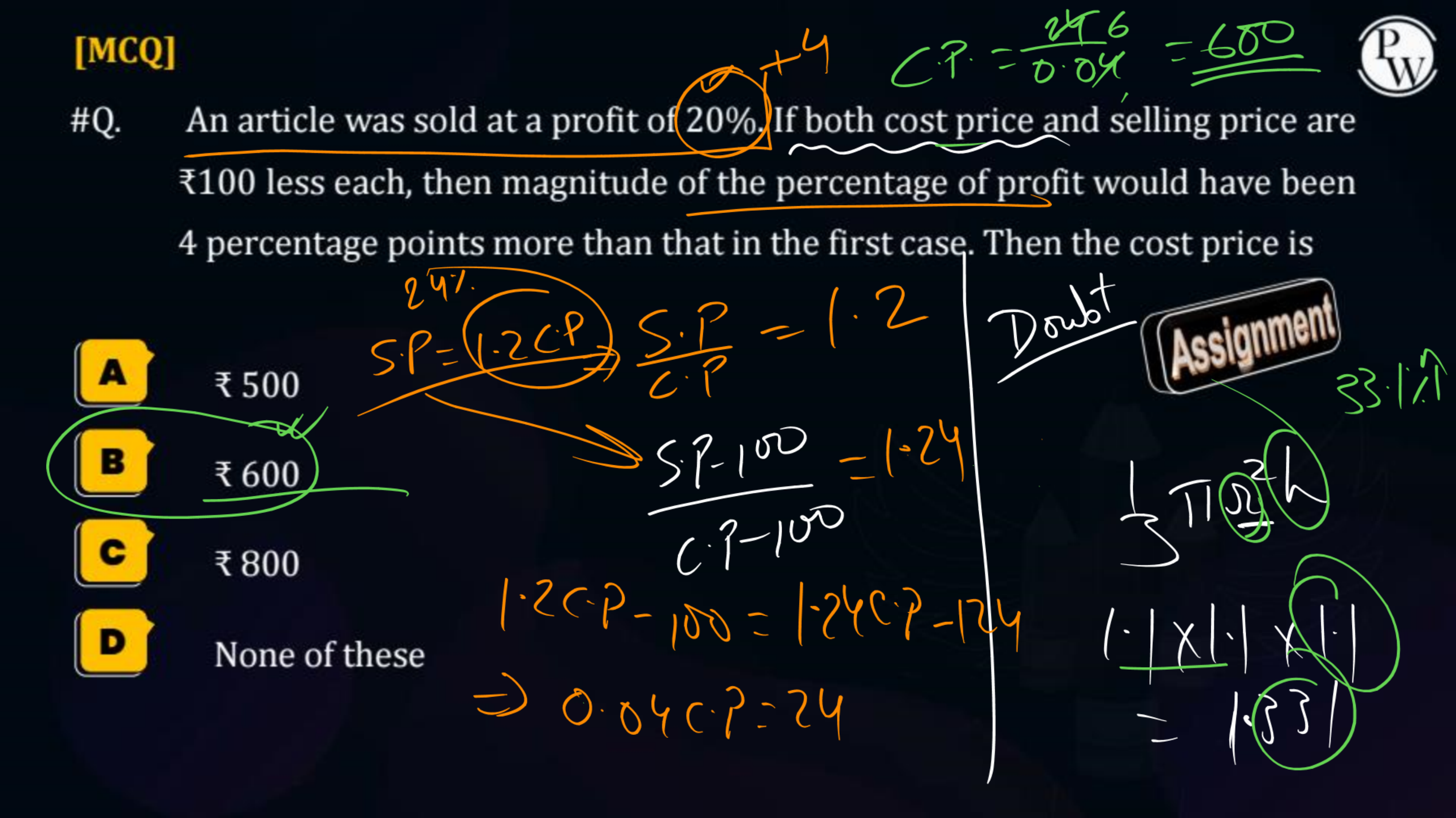
#Q. An article was sold at a profit of 20%. If both cost price and selling price are ₹100 less each, then magnitude of the percentage of profit would have been 4 percentage points more than that in the first case. Then the cost price is

A ₹ 500

B ₹ 600

C ₹ 800

D None of these



[MCQ]

#Q. 5kg of ghee was bought by Vinod for ₹300. One kg from spoilt. He sells the remaining in such a way that on the whole he incurs a loss of 10%. At what price per kg does he sell the ghee?

A ₹ 46.25

B ₹ 45.70

C ₹ 67.50

D ₹ 46.60

$$\frac{S.P}{C.P} = \frac{4}{5} \times \frac{x}{60} = 0.9$$

Assignment

$$\Rightarrow 4x = 270$$

$$x = \frac{270}{4} = 67.5$$

Puzzle:

$P + Q + R = 280$

$40P + Q + 0.2R = 280$







- $P - 40$
- $Q - 1$
- $R - 0.2$

TOTAL Money	:	280/-
TOTAL Number of Product	:	280

$P + Q + R = 40P + Q + 0.2R$

$\frac{P}{R} = \frac{0.8}{39} = \frac{8}{390}$

$39P = 0.8R$

4

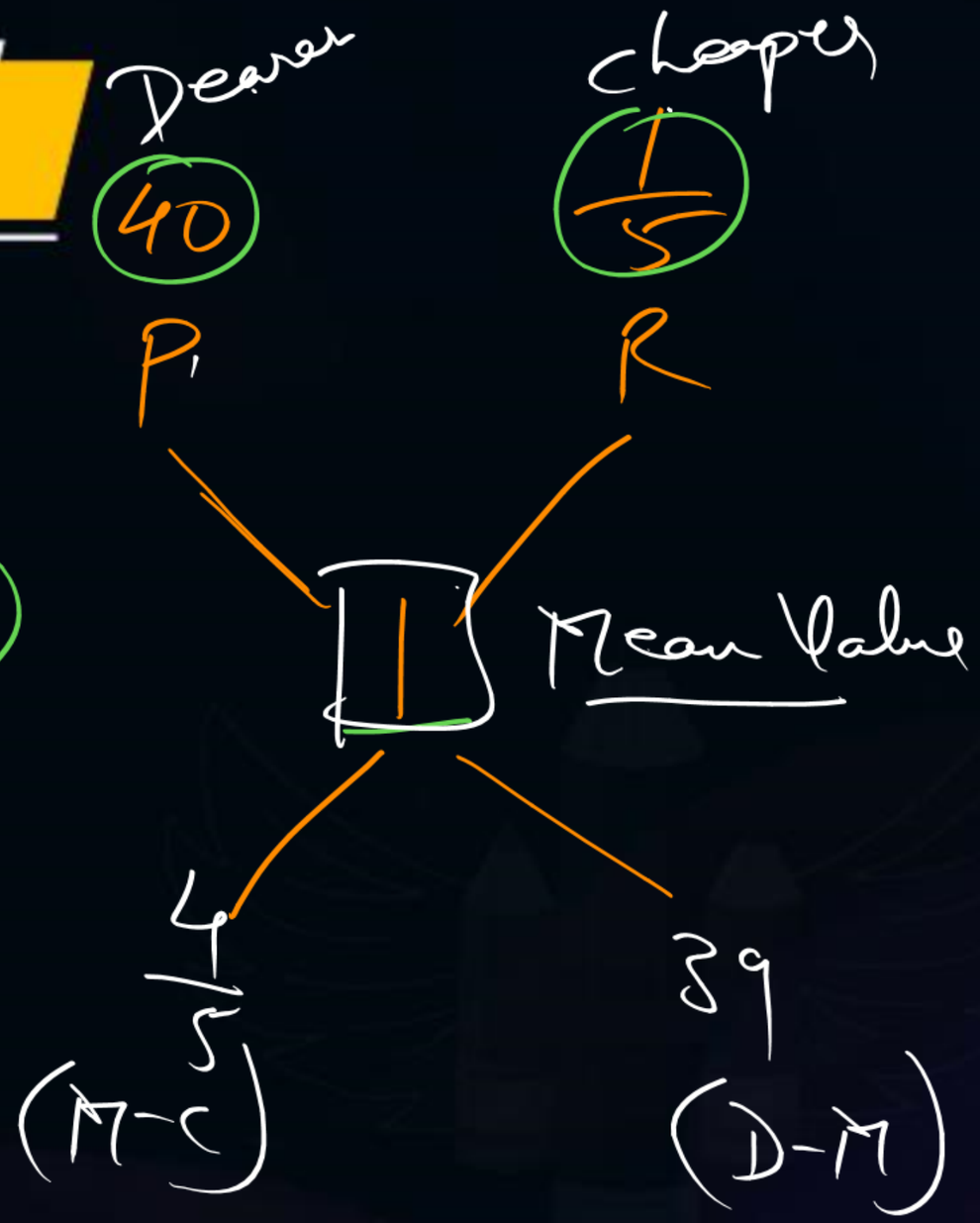
195

No.	Qty
$P \Rightarrow 4$	$\Leftrightarrow 160$
$Q \Rightarrow 81$	$\Leftrightarrow 81$
$R \Rightarrow 195$	$\Leftrightarrow 39$

Mixtures & Alligations

$$P : R = \frac{4}{5} : 39$$

$$P : P = 4 : 195$$



[MCQ]

A : B

#Q. A trader purchases two varieties of rice 'A' & 'B' at the rate of ₹48/kg and ₹60/kg respectively. In what ratio he should mix the two varieties, so that the mixture cost becomes ₹52/kg?

$48A + 60B = 52A + 52B$

$8B = 4A$

$\frac{A}{B} = \frac{8}{4} = \frac{2}{1}$

$A : B = 2 : 1$

$A : B = 8 : 4$

$= 2 : 1$



[MCQ]

#Q. A trader purchases two varieties of Sugar 'P' & 'Q' at the rate of ₹42/kg and ₹80/kg respectively. In what ratio he should ~~mix~~ he two varieties, so that he can sell the mixture at ₹78/kg by getting 30% profit?

S.P = ₹78

$\frac{78}{1.3} = 60$

$\frac{78}{1.3} = 1.3$

P : Q = 20 : 18

= 10 : 9



[MCQ]

$360 \times 20 = 7200$

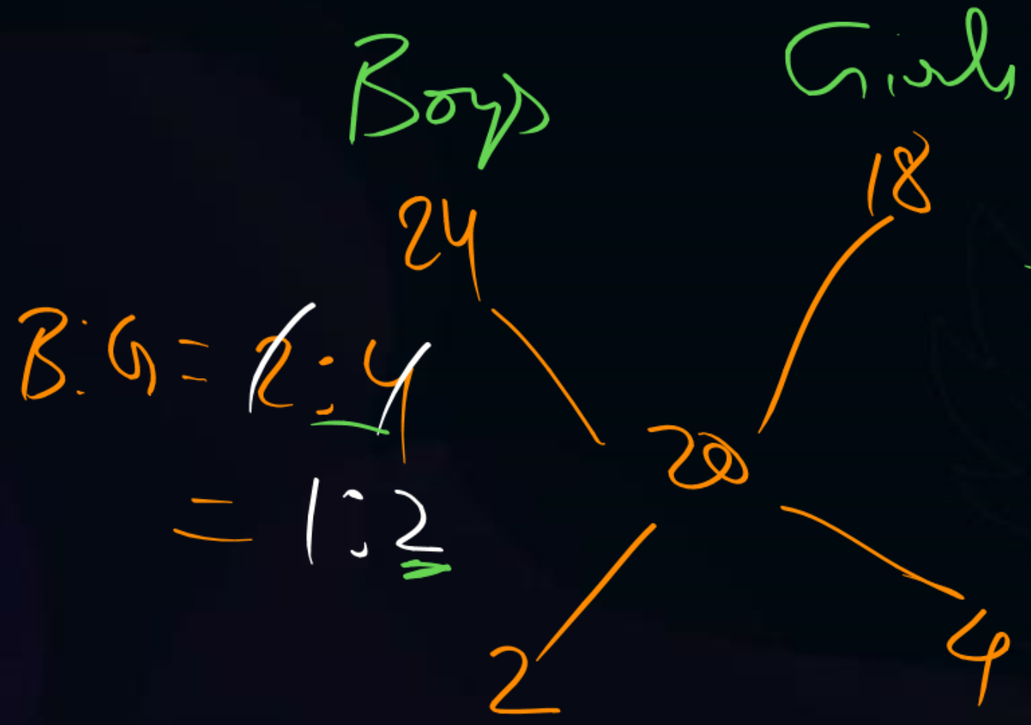
#Q. In a class with 360 students, the average age of all the boys is 24 years whereas average age of all the girls is 18 years. If average age of whole class is 20 years, then find the number of girls in that class.

A 300

B 40

C 120

D 240



$B:G = 2:4$

$= 1:2$

$\frac{2}{3} \times 360^{120}$

$= 240$

$B = 24 \times 120 = 2880$

$G = 18 \times 240 = 4320$

$\frac{4320}{216}$

$= 20$

[MCQ]

#Q. A merchant sells 1600 horses at an overall profit of 25%. Some of them he sold at 20% profit whereas remaining at 40% profit. Find the number of horses he sold at 40% profit.

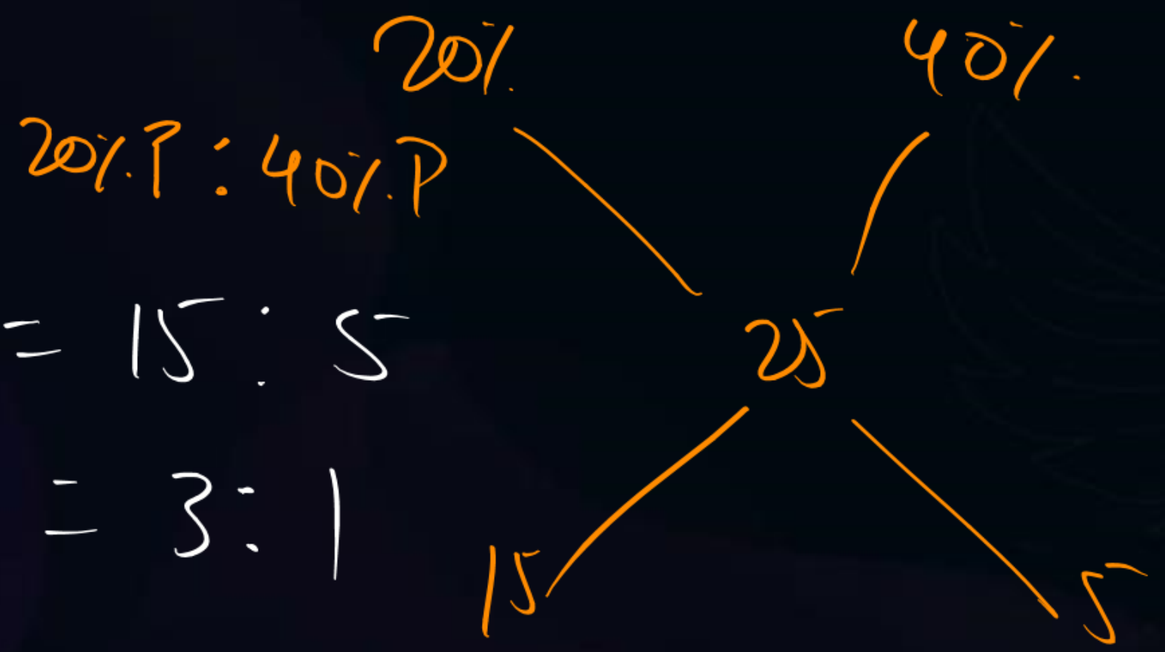
$\frac{1}{4} \times 1600 = 400$

A. 400

B. 1200

C. 1240

D. 300



[MCQ]

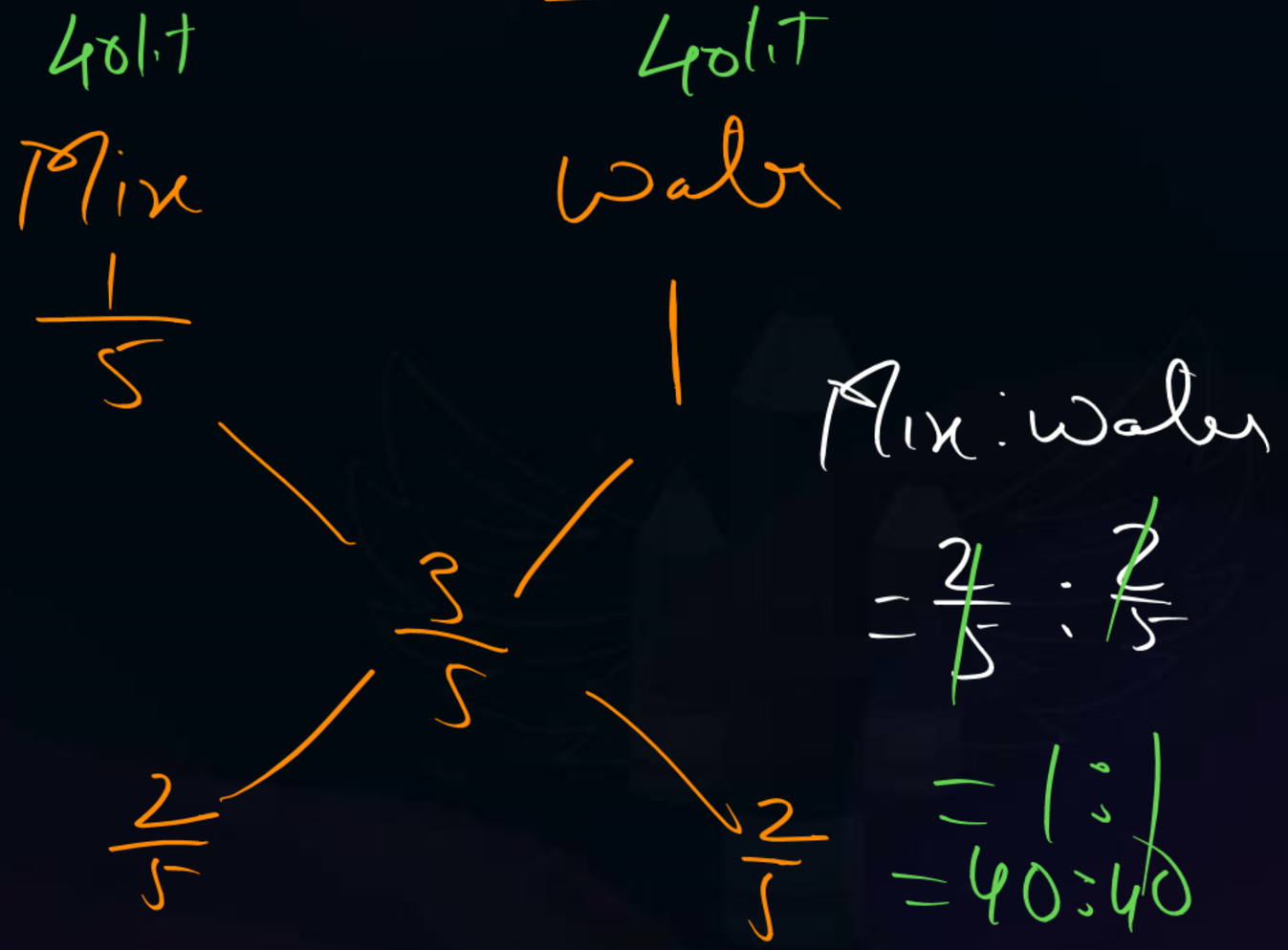
#Q. In a mixture of 40 litres, the ratio of milk and water is 4:1. How much water much be added to this mixture so that the ratio of milk and water becomes 2:3?

A. 20 litres

B. 30 litres

C. 32 litres

D. 40 litres



[MCQ]

#Q. An alloy A consist of iron & copper in the ratio 5 : 7, whereas alloy B consist iron & copper in the ratio 3 : 1. In what ratio these two alloys are to be mixed to form a new alloy with iron & copper in the ratio 2 : 1?

A 5:13

B 7:5

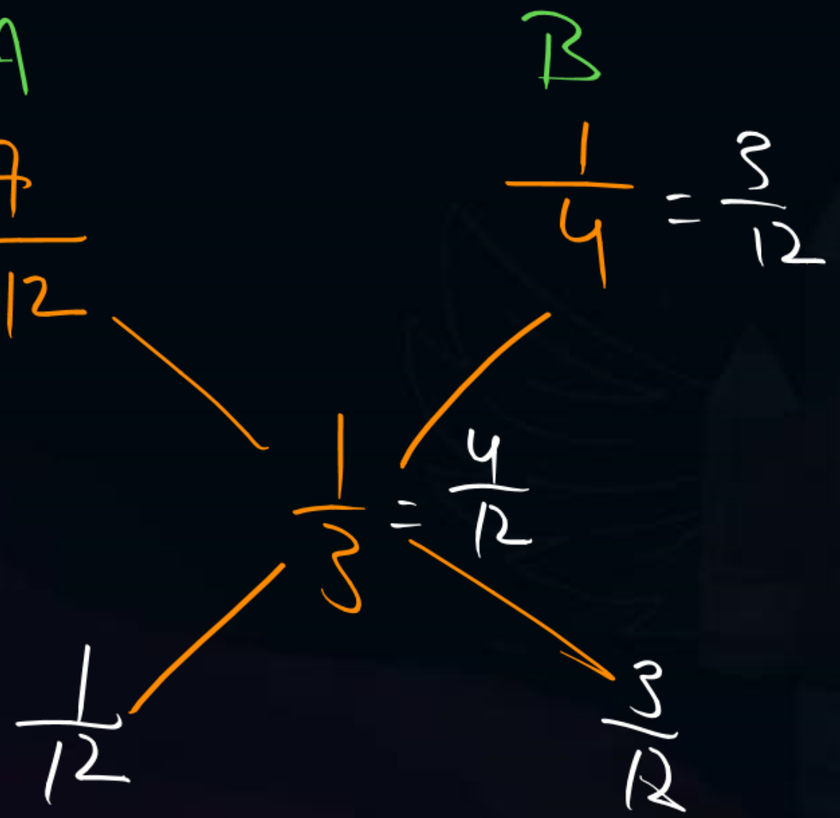
C 1:3

D 3:1

$A:B$

$= \frac{1}{12} : \frac{3}{12}$

$= 1:3$



[MCQ]

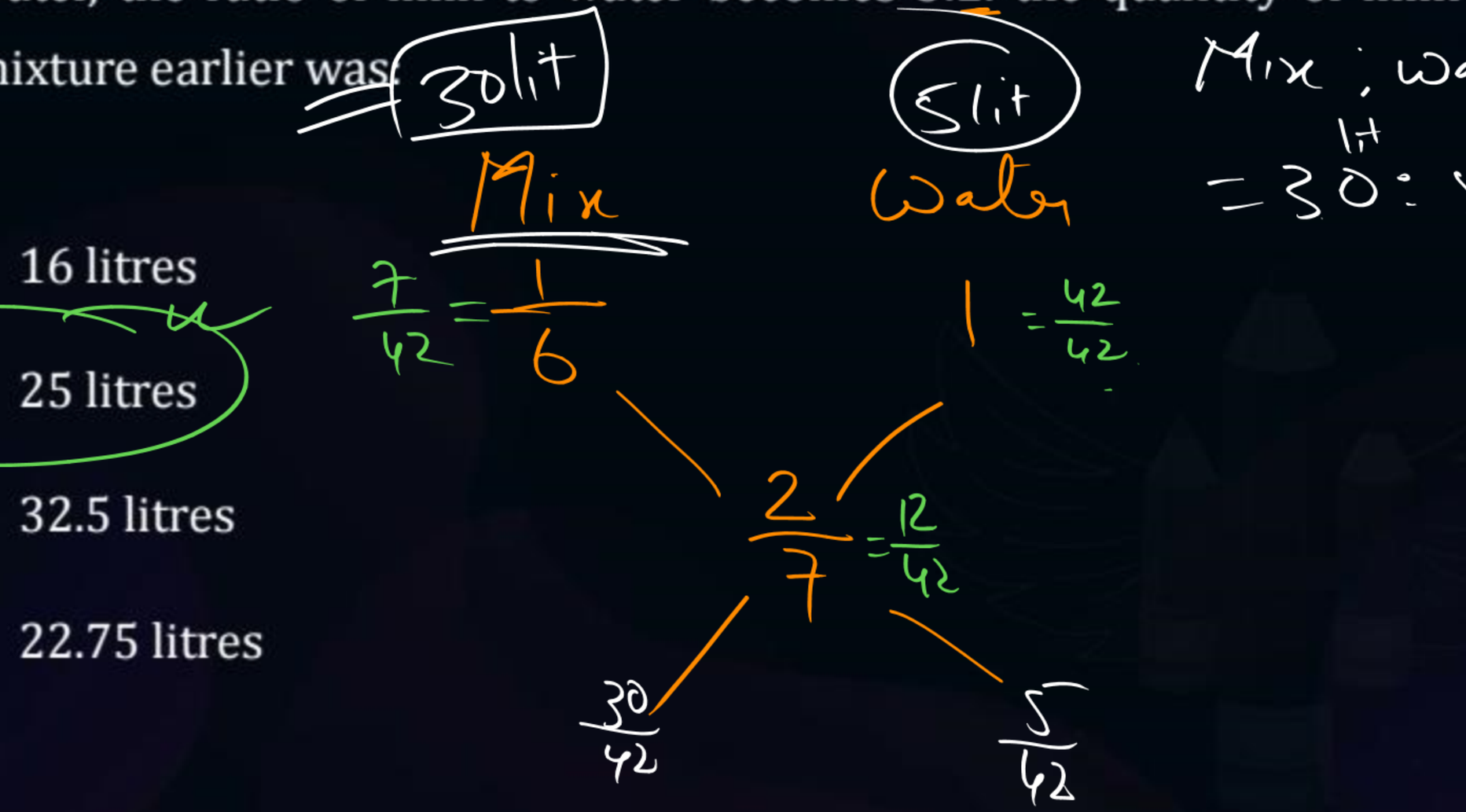
#Q. A mixture contains milk and water in the ratio 5:1. On adding 5 liters of water, the ratio of milk to water becomes 5:2. the quantity of milk in the mixture earlier was:

A. 16 litres

B. 25 litres

C. 32.5 litres

D. 22.75 litres



[MCQ]

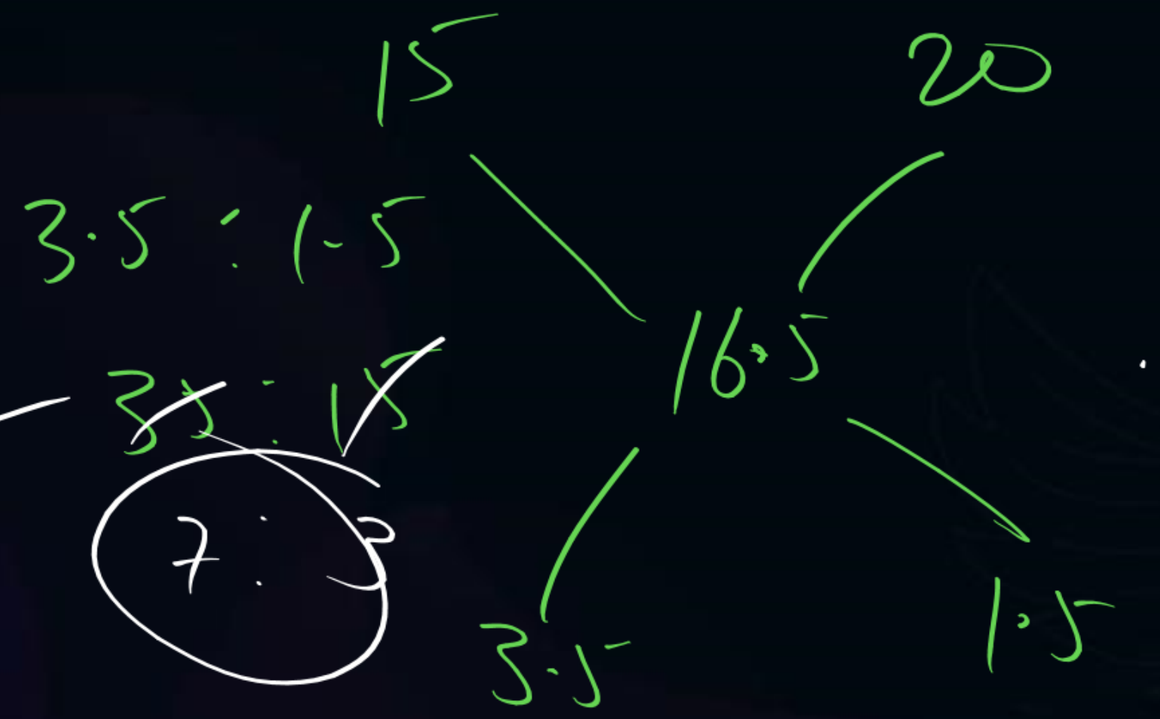
#Q. In what ratio must a grocer mix two varieties of pulses costing ₹15 and ₹20 per kg respectively so as to get a mixture worth ₹16.50 per kg?

A 3:8

B 5:3

C 7:3

D 4:5



[MCQ]

#Q. In what ratio must Ankit mix two varieties of sugar worth ₹20 per kg and ₹32 per kg so that by selling the mixture at ₹36 per kg he may gain 20%?

Assignment

A 1:5

B 3:5

C 2:3

D 3:2

Basics of Number System

$\frac{8}{0} = \infty$

Natural ∞

Even

1, 2, 3, 4, 5, 6, 7, ... — —

0)8(

2)6(3
6

$\frac{6}{2} = 3$

$\frac{16}{2} = 8$

$\frac{15}{3} = 5$

$\frac{0}{9} = 0$

2)16(8
16

3)15(5
15

9)0(

Prime

1 — 1



2

1

Composite

OR

Non-Prime

0 → Even

factors

6 → 1, 2, 3, 6

Multiples

6 → 6, 12, 18, 24, 30, 36

42, 48, 54, 60, 66, 72

78, 84, 90, 96, 102, 108

Rational

$9\overline{)382}\ (42.444$

36

22

18

40

36

40

$6 = \frac{6}{1}$

$3.5 = \frac{35}{10}$

$-8 = \frac{-8}{1}$

$0 = \frac{0}{1}$

$42.\overline{4}$

$\frac{382}{9}$

Lucknow 612 ka

613

Irrational

$\sqrt{2} = 1.4142135623859 \ldots$

Irrational

π

$\frac{22}{7}$ or 3.14

approx Rational

Decimals

Terminating

Non-terminating

Recurring

Non-Recurring

$2\pi r$



2 mins Summary

Topic Mixtures Alligations & Numbers

Basics

THANK - YOU